

Transformer from Scratch

深度学习课程 Project | PyTorch Encoder-Decoder Transformer

核心模块

- Input Embedding
- Positional Encoding
- Scaled Dot-Product Attention
- Multi-Head Attention
- Encoder / Decoder / Mask

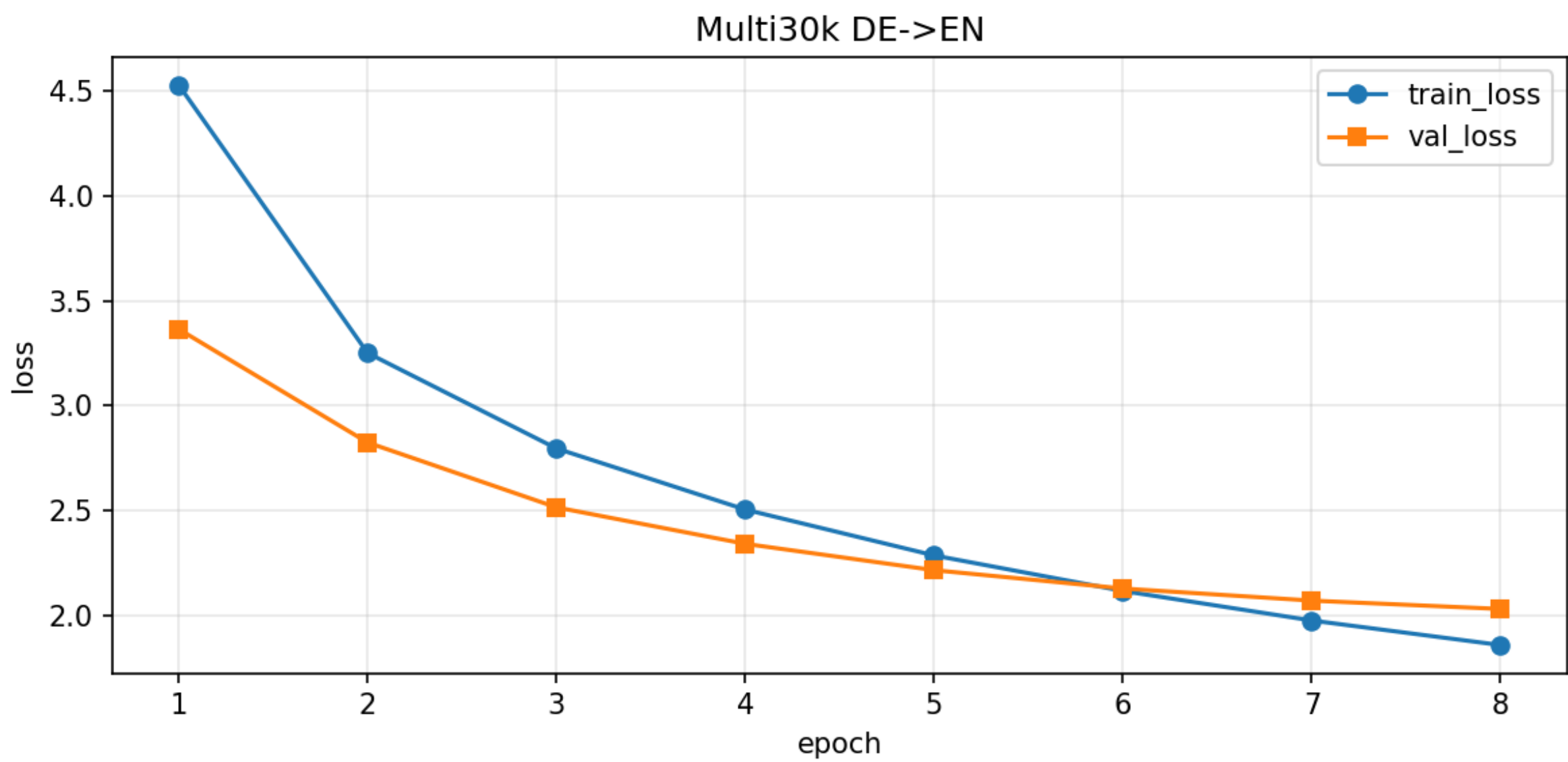
实验设置

- Multi30k / 合成语料
- d_model=256, heads=8, layers=3
- Adam + NLLLoss + grad clip
- Teacher Forcing

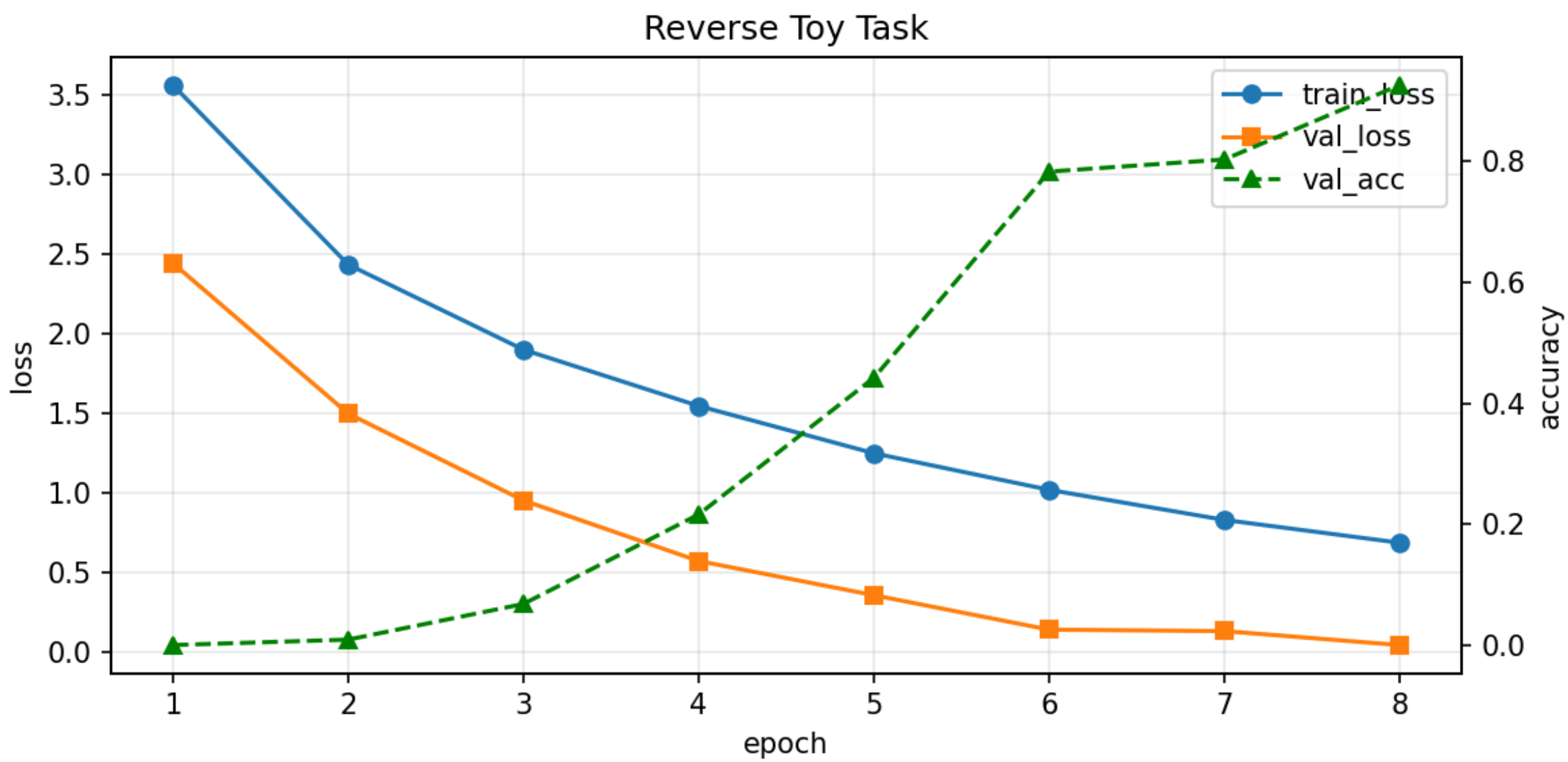
实验结果

- Best val loss: 2.0334
- BLEU: 76.52
- Reverse val acc: 92.58%
- Total params: ~9.88M

模型流程: $\text{src} \rightarrow \text{Embedding} + \text{PE} \rightarrow \text{Encoder} \times N \rightarrow \text{memory} \rightarrow \text{Decoder} \times N \rightarrow \text{Generator} \rightarrow \text{log_softmax}$



翻译任务 Loss 曲线



序列反转任务 Loss 曲线